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(57) Abstract:

The present invention discloses an intelligent emergency response system integrating multi-sensor hazard detection, distributed processing, real-time communication, and controlled mechanical intervention. The system comprises gas and proximity sensing modules, a dual-controller architecture for sensor fusion and decision-making, a GPS-based positioning module, and a communication subsystem capable of transmitting alerts through GSM and cloud platforms. Upon detection of hazardous conditions, the system executes a hierarchical response including local alerts, remote notifications, and activation of a relay-controlled electromechanical actuator configured to perform a controlled glass-breaking operation for emergency escape. The decision-making process is based on multi-parameter evaluation to enhance reliability and minimize false triggering. A power management unit provides regulated multi-level power to all components. The integrated architecture enables autonomous detection, real-time tracking, and adaptive emergency response, making the system suitable for deployment in vehicles, industrial environments, and safety-critical applications.

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COMPLETE SPECIFICATION

(See section 10 and rule13)

TITLE OF THE INVENTION

Adaptive Emergency Response System with Sensor Fusion and Controlled Glass Breaking Mechanism

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Preamble to the description

The following complete specification particularly describes the invention and the manner in which it is to be performed:

Field of the Invention

The present invention relates to the field of intelligent safety systems and emergency response technologies, and more particularly to an integrated, self-powered, multi-sensor-based emergency intervention system configured for real-time hazard detection, predictive risk analysis, location tracking, and controlled mechanical actuation for emergency escape.

The invention further pertains to embedded systems, Internet of Things (IoT)-enabled monitoring platforms, adaptive control systems, and electro-mechanical actuation mechanisms, specifically designed for deployment in vehicles, industrial environments, hazardous zones, and confined spaces requiring rapid and autonomous emergency response.

Background of the Invention

In recent years, there has been a significant increase in the deployment of sensor-based safety systems in vehicles, industrial plants, and smart infrastructure. Conventional safety systems typically rely on single-point sensing mechanisms, such as gas sensors or vibration detectors, which trigger alerts when predefined thresholds are exceeded. However, such systems suffer from several limitations including false positives, delayed response, lack of contextual awareness, and absence of integrated actuation mechanisms.

Existing emergency escape tools, such as manual glass breakers, require human intervention and are often ineffective in critical situations where occupants are incapacitated due to hazardous environmental conditions such as gas leakage, fire, or impact events. Similarly, GPS-based tracking systems and GSM-enabled alert mechanisms operate as standalone modules without providing coordinated decision-making or automated physical intervention.

Further, current IoT-based safety solutions primarily focus on data acquisition and remote monitoring, with limited emphasis on real-time decision intelligence, predictive hazard detection, and synchronized actuation control. The absence of multi-sensor fusion and adaptive control algorithms restricts the reliability and responsiveness of such systems in dynamic and high-risk environments.

Additionally, conventional systems are dependent on continuous external power sources, making them vulnerable to power failures during critical events. There exists a need for a self-powered or energy-optimized system capable of maintaining operation under adverse conditions.

Therefore, there is a need for an integrated, intelligent, and autonomous emergency response system that overcomes the aforementioned limitations by combining multi-sensor data fusion, predictive hazard analysis, adaptive decision-making, controlled mechanical actuation, and real-time communication, thereby enabling rapid and reliable emergency intervention without human dependency.

Objects of the Invention

The primary objective of the present invention is to provide an intelligent emergency response system capable of detecting hazardous conditions through multi-sensor fusion and predictive analysis and initiating appropriate response actions in real time.

Another objective of the invention is to develop a controlled glass-breaking mechanism driven by an adaptive actuation system configured to generate optimized impact force based on situational parameters, thereby ensuring effective emergency escape while minimizing unintended activation.

A further objective of the invention is to incorporate a self-powered or energy-optimized power management subsystem, capable of sustaining system operation through energy harvesting and efficient power regulation techniques during critical scenarios.

Another objective of the invention is to enable real-time location tracking and remote monitoring through integrated GPS and communication modules, facilitating timely emergency response and situational awareness.

Yet another objective of the invention is to implement an intelligent decision-making engine configured to process data from multiple sensors, perform context-aware analysis, and execute hierarchical response strategies including alerts, notifications, and actuation.

A further objective of the invention is to provide a fail-safe and redundant system architecture with distributed processing units to enhance system reliability, responsiveness, and operational robustness in dynamic environments.

Summary of the Invention

The present invention discloses an intelligent, self-powered, and integrated emergency response system configured to perform real-time hazard detection, predictive risk evaluation, and controlled mechanical intervention. The system is designed to operate as a multi-layer safety architecture combining sensing, decision-making, communication, and actuation within a unified framework to enable autonomous emergency handling in critical environments.

The invention incorporates a plurality of environmental and event-detection sensors, including gas sensing and proximity detection modules, which continuously monitor surrounding conditions and generate real-time data streams. These sensor inputs are processed through a distributed control architecture comprising a primary processing unit and a secondary interface controller, wherein the primary unit executes predictive hazard analysis using predefined and adaptive decision logic, and the secondary unit manages peripheral interfacing and local system responses. The distributed architecture ensures optimized computational load sharing and enhances system responsiveness and reliability.

The system further integrates a real-time positioning module configured to acquire geospatial coordinates, which are transmitted through a communication subsystem to remote platforms, thereby enabling continuous tracking and situational awareness. The communication subsystem is configured to transmit alerts through short message services and cloud-based interfaces, ensuring multi-channel notification and redundancy in emergency communication.

A key feature of the invention is the provision of a controlled mechanical actuation subsystem configured to perform a glass-breaking operation in response to validated emergency conditions. The actuation subsystem is driven by an adaptive control mechanism that regulates the operational parameters of an electromechanical actuator to deliver sufficient impact force for effective glass breakage while minimizing the risk of unintended activation. The activation logic is governed by a multi-parameter decision engine that evaluates sensor inputs, environmental context, and system thresholds prior to actuation.

The invention further incorporates a power management subsystem configured to provide regulated multi-level power outputs for system components, wherein energy optimization techniques are employed to ensure sustained operation during power interruptions. The architecture is adaptable to include auxiliary energy support mechanisms, thereby enhancing system autonomy in critical scenarios.

The system additionally provides user interfacing through a display module and alerting devices to indicate system status and emergency conditions locally, while maintaining synchronization with remote monitoring platforms. The integrated design ensures coordinated operation of sensing, processing, communication, and actuation subsystems, thereby achieving a comprehensive and autonomous emergency response solution.

Brief Description of the Drawings

Figure 1 illustrates the hardware configuration of the intelligent emergency response system, showing the integration of sensing modules, control units, communication components, power supply elements, and the electromechanical actuation subsystem mounted on a structural frame.

Figure 2 illustrates the circuit diagram of the system, depicting the interconnection between the primary and secondary microcontroller units, sensor modules, communication interfaces, relay-based switching circuitry, power regulation stages, and actuator control pathways.

Figure 3 illustrates the remote monitoring and tracking interface, showing the visualization of real-time geospatial data and system status on a cloud-based dashboard, enabling remote access and continuous situational awareness.

Figure 4 illustrates the alert notification mechanism of the system, showing the transmission of emergency alerts, including vibration detection and gas leakage notifications, through communication channels to designated users.

Detailed Description of the Invention

The present invention relates to an intelligent, integrated emergency response system configured to detect hazardous conditions, perform context-aware decision-making, and initiate controlled mechanical intervention along with real-time communication and tracking. The system is realized as a compact, modular assembly comprising sensing units, distributed processing units, communication modules, an electromechanical actuation mechanism, and a regulated power management subsystem, all mounted on a structural support frame as illustrated in Figure 1 .

The system includes a plurality of sensing elements configured to continuously monitor environmental and situational parameters. The sensing subsystem comprises a gas sensor for detecting hazardous gas concentrations and an infrared sensor for detecting proximity or motion-based events. These sensors generate analog or digital signals corresponding to environmental conditions, which are periodically sampled and conditioned through the processing units. Additional input is provided through a manual interface element in the form of a push button, enabling user-triggered activation during emergency scenarios.

A distributed control architecture is employed, consisting of a primary processing unit and a secondary processing unit, both implemented using microcontroller platforms. The primary processing unit is configured to perform core system operations including acquisition of sensor data, execution of decision logic, communication handling, and control of actuation sequences. The secondary processing unit is configured to manage peripheral interfacing, including display control, local alert generation, and preprocessing of sensor signals where required. This distributed configuration reduces computational latency, enhances system responsiveness, and provides redundancy in critical operations.

The decision-making process is governed by a multi-parameter evaluation mechanism implemented within the primary processing unit. Sensor inputs from the gas sensor and infrared sensor are continuously monitored and compared against predefined threshold values. In addition to threshold-based triggering, the system incorporates a context-aware evaluation logic wherein multiple sensor conditions are evaluated in combination to validate the occurrence of an emergency event. For instance, a gas leakage condition may be confirmed based on sustained sensor readings exceeding a threshold over a defined duration, optionally combined with motion inactivity or abnormal environmental patterns. This multi-sensor fusion approach minimizes false triggering and enhances reliability.

Upon detection of a validated emergency condition, the system initiates a hierarchical response sequence. The first level of response involves activation of a local alerting mechanism, including a buzzer and a display module. The display unit, interfaced through an I2C communication module, presents real-time system status, including sensor readings and location data. Simultaneously, the communication subsystem is activated to transmit alerts to remote users.

The communication subsystem comprises a GSM module configured to send short message service notifications to predefined recipients. The alert messages include information indicative of the detected hazard condition. In parallel, the system interfaces with a cloud-based platform through a wireless communication module, enabling remote visualization of system parameters and status, as illustrated in Figure 3 . The integration of both GSM-based and cloud-based communication ensures redundancy and reliability in information dissemination.

The system further incorporates a global positioning module configured to continuously acquire geographic coordinates in terms of latitude and longitude. These coordinates are processed by the primary controller and transmitted along with alert messages, enabling precise location tracking of the system in real time. The tracking functionality supports emergency response coordination and remote monitoring.

A key functional component of the invention is the electromechanical actuation subsystem configured to perform a glass-breaking operation during emergency conditions. The actuation subsystem comprises a relay-controlled switching mechanism interfaced with a high-power actuator, such as a permanent magnet direct current motor. The relay module provides electrical isolation between the low-power control circuitry and the high-power actuation circuit. Upon receiving an actuation command from the primary controller, the relay is energized, thereby activating the actuator.

The actuation process is governed by a controlled triggering mechanism wherein the system ensures that the glass-breaking operation is executed only after confirmation of emergency conditions through the multi-parameter decision logic. The actuator is driven for a controlled duration and operational intensity sufficient to generate the required mechanical force for breaking glass structures. The controlled operation reduces the likelihood of accidental or unintended actuation, thereby enhancing system safety.

The power management subsystem is configured to provide stable and regulated power to all system components. The system receives input power through a step-down transformer, which converts high-voltage alternating current to a lower voltage level. This is followed by a rectification stage to convert alternating current to direct current, and a filtering stage to smooth the output. Voltage regulation circuits are employed to provide stable voltage levels required by different subsystems, including 5V and 3.3V outputs for microcontrollers and sensors. A DC-DC conversion stage is incorporated to supply higher voltage levels required for the actuation subsystem. The power architecture ensures uninterrupted operation of sensing, processing, and communication components.

The system is further designed to support energy optimization through efficient power distribution and selective activation of subsystems. Non-critical components are maintained in low-power states during normal operation and are activated only upon detection of relevant events. This approach enhances overall energy efficiency and supports extended operational capability during constrained power conditions.

The hardware configuration, as illustrated in Figure 1 and Figure 2 , demonstrates the integration of sensing modules, dual microcontroller units, communication interfaces, power circuitry, and actuation components within a unified assembly. The alerting mechanism illustrated in Figure 4 shows the transmission of emergency notifications corresponding to detected events such as gas leakage and vibration.

In operation, the system continuously monitors environmental conditions, evaluates sensor data using context-aware logic, and triggers a sequence of alerts and actuation responses upon detection of hazardous conditions. The integration of sensing, intelligent processing, communication, and controlled actuation within a single system enables rapid, reliable, and autonomous emergency response, thereby significantly improving safety in critical environments.

Claims

We claim:

1. An intelligent emergency response system comprising a plurality of sensing modules, a distributed processing architecture, a communication subsystem, a positioning module, an electromechanical actuation subsystem, and a power management unit, wherein the sensing modules are configured to detect environmental and situational parameters including gas concentration and proximity, the distributed processing architecture comprising a primary controller and a secondary controller configured to acquire and process sensor data, execute a multi-parameter decision logic based on sensor fusion and contextual evaluation, and generate control signals, the communication subsystem configured to transmit alerts through wireless communication and short message services, the positioning module configured to provide real-time geographic coordinates, the electromechanical actuation subsystem comprising a relay-controlled actuator configured to perform a controlled glass-breaking operation upon validation of an emergency condition, and the power management unit configured to provide regulated power to the system components, wherein the system autonomously detects hazardous conditions, performs hierarchical response actions, and enables real-time tracking and controlled emergency intervention.

2. The system as claimed in claim 1, wherein the multi-parameter decision logic is configured to evaluate sensor inputs based on threshold comparison, temporal persistence of detected conditions, and combination of multiple sensor states to validate emergency scenarios and minimize false triggering.

3. The system as claimed in claim 1, wherein the electromechanical actuation subsystem is configured to operate in a controlled manner by regulating actuation duration and activation conditions, thereby ensuring sufficient mechanical force for glass breaking while preventing unintended or accidental activation.

4. The system as claimed in claim 1, wherein the communication subsystem is configured to transmit alert notifications simultaneously through a cloud-based remote monitoring platform and a GSM-based messaging system, thereby providing redundant and reliable communication of emergency events along with location data.

5. The system as claimed in claim 1, wherein the power management unit comprises a step-down transformer, rectification and filtering stages, voltage regulation circuitry, and a DC-DC conversion module configured to supply multiple voltage levels and enable energy-optimized operation of sensing, processing, communication, and actuation subsystems.

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Abstract

Adaptive Emergency Response System with Sensor Fusion and Controlled Glass Breaking Mechanism

The present invention discloses an intelligent emergency response system integrating multi-sensor hazard detection, distributed processing, real-time communication, and controlled mechanical intervention. The system comprises gas and proximity sensing modules, a dual-controller architecture for sensor fusion and decision-making, a GPS-based positioning module, and a communication subsystem capable of transmitting alerts through GSM and cloud platforms. Upon detection of hazardous conditions, the system executes a hierarchical response including local alerts, remote notifications, and activation of a relay-controlled electromechanical actuator configured to perform a controlled glass-breaking operation for emergency escape. The decision-making process is based on multi-parameter evaluation to enhance reliability and minimize false triggering. A power management unit provides regulated multi-level power to all components. The integrated architecture enables autonomous detection, real-time tracking, and adaptive emergency response, making the system suitable for deployment in vehicles, industrial environments, and safety-critical applications.

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